

Cloud

47 questions · compiled from 1 file

CLOUD CHARACTERISTICS

1. Which cloud characteristic best explains why a customer can provision a virtual server without speaking to the provider?

1. Resource pooling
2. On-demand self-service
3. Measured service
4. Multi-tenancy

Answer 2 · On-demand self-service

CLOUD CHARACTERISTICS

2. A company is billed based on storage consumed, API calls made, and compute hours used. Which cloud characteristic does this reflect?

1. Rapid elasticity
2. Measured service
3. Fault isolation
4. Virtualization

Answer 2 · Measured service

SERVICE MODELS

3. Which option best describes PaaS?

1. A complete end-user application delivered over the web
2. A managed platform for building and deploying applications
3. Raw compute, storage, and networking resources
4. A physical data center leased monthly

Answer 2 · A managed platform for building and deploying applications

SERVICE MODELS

4. Which scenario is the clearest example of SaaS?

1. A developer launches a Linux VM
2. A team uses a browser-based CRM managed by the vendor
3. An engineer configures a VPC and subnets
4. A company installs Kubernetes on self-managed servers

Answer 2 · A team uses a browser-based CRM managed by the vendor

DEPLOYMENT MODELS

5. Which deployment model is most appropriate when an organization must keep some regulated workloads on-premises while extending burst capacity to the cloud?

1. Public cloud
2. Private cloud
3. Hybrid cloud
4. Community cloud

Answer 3 · Hybrid cloud

6. What is the most accurate distinction between elasticity and scalability?

1. Elasticity is manual and scalability is automatic
2. Elasticity adjusts resources with changing demand, while scalability is the ability to support growth
3. Scalability applies only to storage, elasticity only to compute
4. They are interchangeable terms

Answer 2 · Elasticity adjusts resources with changing demand, while scalability is the ability to support growth

7. An application automatically adds instances during traffic spikes and removes them when demand falls. This is primarily an example of what?

1. Vertical scaling only
2. Elasticity
3. Cold storage
4. Static provisioning

Answer 2 · Elasticity

8. Why is OpEx often associated with cloud adoption?

1. Because cloud always eliminates monthly costs
2. Because cloud shifts spending from large upfront purchases to ongoing usage-based expenses
3. Because OpEx means buying more servers in advance
4. Because OpEx applies only to networking

Answer 2 · Because cloud shifts spending from large upfront purchases to ongoing usage-based expenses

9. Why do AWS Regions contain multiple Availability Zones?

1. To increase IAM policy size
2. To support resilience and fault tolerance
3. To reduce the need for DNS
4. To replace VPCs

Answer 2 · To support resilience and fault tolerance

10. Which design best improves resilience for a production web application?

1. One EC2 instance in one Availability Zone
2. Two EC2 instances in different Availability Zones behind a load balancer
3. One S3 bucket with no redundancy planning
4. One Lambda function with no monitoring

Answer 2 · Two EC2 instances in different Availability Zones behind a load balancer

11. Which AWS service is most appropriate when you need full operating system control over a long-running application server?

1. AWS Lambda
2. Amazon EC2
3. Amazon SQS
4. Amazon CloudFront

Answer 2 · Amazon EC2

COMPUTE

12. Which workload is best suited to AWS Lambda rather than EC2?

1. A database server running continuously
2. An event-driven image resize function triggered on file upload
3. A legacy application requiring persistent local configuration
4. A custom OS appliance

Answer 2 · An event-driven image resize function triggered on file upload

STORAGE

13. Which statement best distinguishes Amazon S3 from Amazon EBS?

1. S3 is block storage and EBS is object storage
2. S3 is object storage and EBS is block storage
3. Both are object storage services
4. Both are attached only to Lambda

Answer 2 · S3 is object storage and EBS is block storage

STORAGE

14. A team needs low-cost archival storage for compliance records that are rarely accessed. Which AWS option is most appropriate?

1. Amazon EBS
2. Amazon S3 Standard
3. Amazon S3 Glacier
4. Instance store

Answer 3 · Amazon S3 Glacier

STORAGE

15. Which storage option is most suitable for a boot volume attached to an EC2 instance?

1. Amazon CloudFront
2. Amazon EBS
3. Amazon Route 53
4. Amazon SNS

Answer 2 · Amazon EBS

CONTENT DELIVERY

16. Why would an architect place CloudFront in front of an S3-hosted static site?

1. To create IAM users
2. To reduce latency by caching content closer to users
3. To convert object storage into block storage
4. To replace DNS

Answer 2 · To reduce latency by caching content closer to users

CONTENT DELIVERY

17. Which statement is most accurate about using CloudFront with S3?

1. CloudFront only works with public S3 buckets
2. CloudFront can serve content from S3 and improve delivery performance
3. CloudFront replaces S3 storage entirely
4. CloudFront is only for database traffic

Answer 2 · CloudFront can serve content from S3 and improve delivery performance

SECURITY

18. Which AWS service is used to manage identities, roles, and permissions?

1. Amazon GuardDuty
2. AWS IAM
3. Amazon CloudWatch
4. AWS Lambda

Answer 2 · AWS IAM

SECURITY

19. What is the best example of the principle of least privilege?

1. Giving all developers administrator access
2. Granting a user only the permissions required for their task
3. Using the root account for daily work
4. Sharing one account across the team

Answer 2 · Granting a user only the permissions required for their task

SECURITY

20. Under the shared responsibility model, which task is generally the customer's responsibility?

1. Securing AWS data center buildings
2. Maintaining the physical network hardware
3. Configuring access to their applications and data
4. Replacing failed disks in AWS facilities

Answer 3 · Configuring access to their applications and data

SECURITY

21. Which practice is most appropriate for the AWS root account?

1. Use it for application deployments
2. Share it with the operations team
3. Protect it with MFA and avoid routine use
4. Attach it to EC2 instances

Answer 3 · Protect it with MFA and avoid routine use

NETWORKING

22. What is the primary purpose of an Amazon VPC?

1. To provide object storage
2. To define a logically isolated virtual network
3. To distribute content globally
4. To queue messages

Answer 2 · To define a logically isolated virtual network

NETWORKING

23. What is a subnet in AWS?

1. A DNS routing policy
2. A segmented portion of a VPC IP range
3. A type of IAM role
4. A storage class

Answer 2 · A segmented portion of a VPC IP range

NETWORKING

24. Which component is required for internet communication from public resources in a VPC?

1. IAM Role
2. Internet Gateway
3. CloudWatch Alarm
4. SQS Queue

Answer 2 · Internet Gateway

NETWORKING

25. Which AWS service is primarily responsible for DNS routing and domain management?

1. Amazon ECS
2. Amazon Route 53
3. Amazon EBS
4. AWS Artifact

Answer 2 · Amazon Route 53

HIGH AVAILABILITY

26. Why is deploying across multiple Availability Zones considered a best practice for critical applications?

1. It reduces the need for IAM
2. It improves fault tolerance if one zone fails
3. It guarantees zero cost
4. It eliminates the need for backups

Answer 2 · It improves fault tolerance if one zone fails

LOAD BALANCING

27. What is the main role of Elastic Load Balancing?

1. Encrypt all S3 objects
2. Distribute incoming traffic across multiple targets
3. Create IAM users automatically
4. Store application logs

Answer 2 · Distribute incoming traffic across multiple targets

LOAD BALANCING

28. Why is Elastic Load Balancing often paired with Auto Scaling?

1. Because ELB stores backups
2. Because together they distribute traffic and adjust capacity to demand
3. Because Auto Scaling replaces DNS
4. Because ELB creates VPCs

Answer 2 · Because together they distribute traffic and adjust capacity to demand

AUTO SCALING

29. What is the primary benefit of Auto Scaling?

1. It manually provisions resources once per month
2. It automatically adjusts compute capacity based on demand
3. It replaces IAM policies
4. It archives old objects

Answer 2 · It automatically adjusts compute capacity based on demand

MESSAGING

30. Which AWS service is best described as a managed message queue for decoupling components?

1. Amazon SNS
2. Amazon SQS
3. Amazon Route 53
4. Amazon CloudFront

Answer 2 · Amazon SQS

MESSAGING

31. Which service is most appropriate for a publish/subscribe pattern where one event should fan out to multiple subscribers?

1. Amazon EBS
2. Amazon SNS
3. Amazon EC2
4. Amazon VPC

Answer 2 · Amazon SNS

MESSAGING

32. Which pairing best supports both fan-out notifications and durable downstream processing?

1. SNS and SQS
2. EC2 and EBS
3. IAM and CloudTrail
4. Route 53 and CloudFront

Answer 1 · SNS and SQS

CONTAINERS

33. Which AWS service is designed to run and orchestrate containers without managing Kubernetes directly?

1. Amazon ECS
2. Amazon Route 53
3. Amazon S3 Glacier
4. AWS IAM

Answer 1 · Amazon ECS

CONTAINERS

34. Which AWS service provides managed Kubernetes?

1. Amazon ECS
2. Amazon EKS
3. AWS Lambda
4. Amazon SQS

Answer 2 · Amazon EKS

CONTAINERS VS SERVERLESS

35. Which statement best compares containers and serverless functions?

1. Lambda provides more OS-level control than containers
2. Containers are useful when you need more runtime control, while Lambda is ideal for event-driven execution
3. Containers cannot scale
4. Lambda requires managing servers directly

Answer 2 · Containers are useful when you need more runtime control, while Lambda is ideal for event-driven execution

MONITORING

36. Which AWS service is used to collect metrics, logs, and alarms?

1. AWS IAM
2. Amazon CloudWatch
3. Amazon S3
4. Amazon Inspector

Answer 2 · Amazon CloudWatch

MONITORING

37. If CPU utilization remains above a threshold for several minutes and triggers a scaling action, which service is watching that metric?

1. Amazon Route 53
2. Amazon CloudWatch
3. Amazon SQS
4. Amazon ECS

Answer 2 · Amazon CloudWatch

STATIC WEBSITE HOSTING

38. Which AWS combination is commonly used to host a static website with improved global performance?

1. EC2 and RDS
2. S3 and CloudFront
3. Lambda and SQS
4. IAM and VPC

Answer 2 · S3 and CloudFront

STATIC WEBSITE HOSTING

39. Why might an architect prefer CloudFront in front of S3 rather than exposing only the S3 website endpoint?

1. To reduce object durability
2. To improve caching and content delivery performance
3. To remove the need for DNS
4. To convert the site into a dynamic backend

Answer 2 · To improve caching and content delivery performance

FAULT TOLERANCE

40. What does fault tolerance mean in cloud architecture?

1. The system can continue operating despite component failures
2. The system never needs monitoring
3. The system avoids all software bugs
4. The system uses only one server

Answer 1 · The system can continue operating despite component failures

ARCHITECTURE DESIGN

41. A web application needs low latency, high availability, and the ability to handle traffic spikes. Which design is strongest?

1. Single EC2 instance in one AZ
2. S3 only with no CDN
3. EC2 in multiple AZs behind ELB with Auto Scaling and CloudFront for static assets
4. One private subnet with no internet path

Answer 3 · EC2 in multiple AZs behind ELB with Auto Scaling and CloudFront for static assets

SHARED RESPONSIBILITY

42. Which statement best reflects the AWS shared responsibility model?

1. AWS manages all customer IAM permissions
2. Customers are responsible only for billing
3. AWS secures the cloud, while customers secure what they run and configure in the cloud
4. Customers secure AWS data centers

Answer 3 · AWS secures the cloud, while customers secure what they run and configure in the cloud

ROUTE 53 AND RESILIENCE

43. How can Route 53 improve application availability?

1. By increasing EC2 memory
2. By routing users to healthy endpoints using DNS policies and health checks
3. By replacing CloudWatch alarms
4. By attaching storage volumes

Answer 2 · By routing users to healthy endpoints using DNS policies and health checks

COST OPTIMIZATION

44. Which design choice most directly supports cost efficiency for variable traffic workloads?

1. Provisioning peak capacity all the time
2. Using Auto Scaling to match resources to demand
3. Running all workloads on the root account
4. Using only one Availability Zone

Answer 2 · Using Auto Scaling to match resources to demand

IAM AND ACCESS CONTROL

45. Why are IAM roles generally preferred over sharing long-term credentials across workloads?

1. Roles reduce the need for networking
2. Roles provide controlled, temporary access without embedding static credentials
3. Roles increase storage durability
4. Roles replace CloudFront

Answer 2 · Roles provide controlled, temporary access without embedding static credentials

CLOUD ADOPTION

46. Which benefit of cloud most directly supports faster experimentation by development teams?

1. Long procurement cycles
2. On-demand provisioning of resources
3. Manual hardware installation
4. Single-region dependency

Answer 2 · On-demand provisioning of resources

RESILIENCE AND PERFORMANCE

47. Which combination best addresses both global performance and regional failure concerns?

1. One EC2 instance and one EBS volume
2. CloudFront plus multi-AZ application deployment
3. One private S3 bucket with no distribution layer
4. One IAM user with admin access

Answer 2 · CloudFront plus multi-AZ application deployment